

Recognizing Emotional States of Dogs from Images Using Deep Learning

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Motivation & Problem

Goal: classify dog emotional states from images

Classes: Angry, Happy, Relaxed, Sad

Applications: pet monitoring, safety, behavior analysis



Challenges

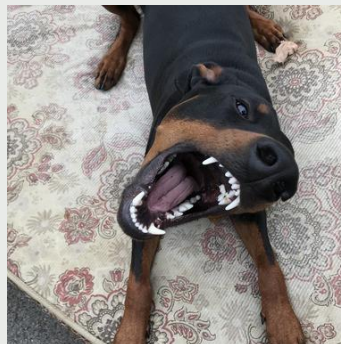
- High visual **variation** across breeds
- **Subtle** emotional cues
- **Overlapping features** between classes
- Dataset **noise** (pose, lighting, background)



Dataset

Kaggle Dataset: <https://www.kaggle.com/danielshanbalico/dog-emotion>

- Comprises of **4000** dog images labeled with the emotions angry, happy, relaxed, and sad
- Each emotion contains **1000** dog images with a comprehensive variety of dog species
- Images range from **strictly dog faces** to **full body images**
- Images vary in **resolution** and **dimensions**



Why Deep Learning?

Why CNNs?

- Images have **spatial structure**
- Emotion cues are **local + global**
- Handcrafted features **insufficient**
- **Hard to generalize** across species, lighting, poses

Examples

- Eyes & mouth → local
- Body posture → global



Architecture

Transfer Learning with CNNs (ResNet family)

Backbones:

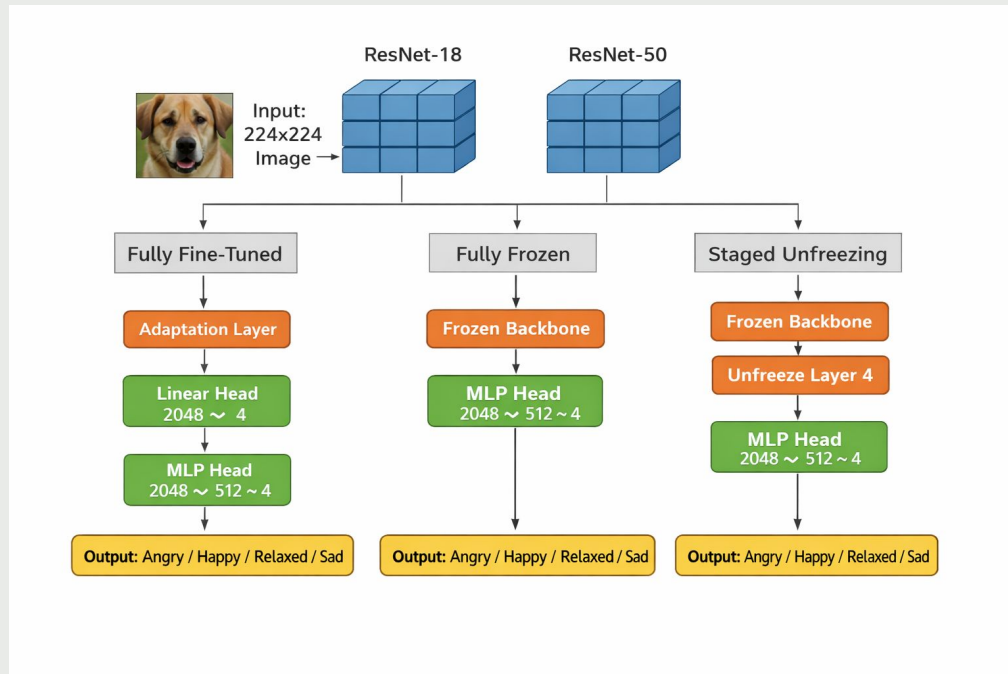
- ResNet-18
- ResNet-50

Classifier heads:

- Linear ($2048 \rightarrow 4$)
- MLP ($2048 \rightarrow 512 \rightarrow 4$)

Training modes:

- Fully fine-tuned
- Fully frozen
- Staged unfreezing (head $\rightarrow$ layer4)



Training

Preprocessing

- Resize (224×224), Normalize
- Augmentation: flip, rotation, zoom, brightness

Training Setup

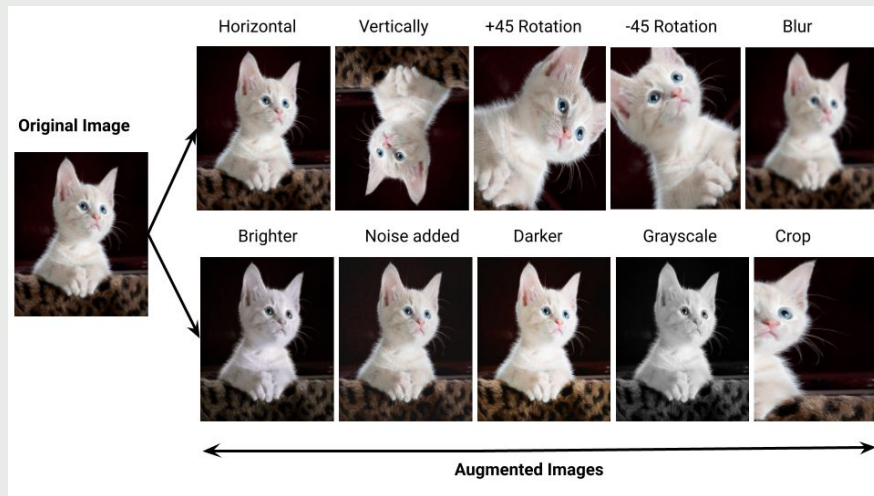
- Adam, LR = $1e-4$
- Batch size = 32, Epochs = 25–40
- Cross-entropy loss

Training Strategies

- Fine-tuned
- Frozen
- Staged unfreezing

Dataset Split

- 70 / 15 / 15



Evaluation

Quantitative:

- Test set evaluation
- Accuracy, Precision, Recall, F1-score
- Confusion matrix

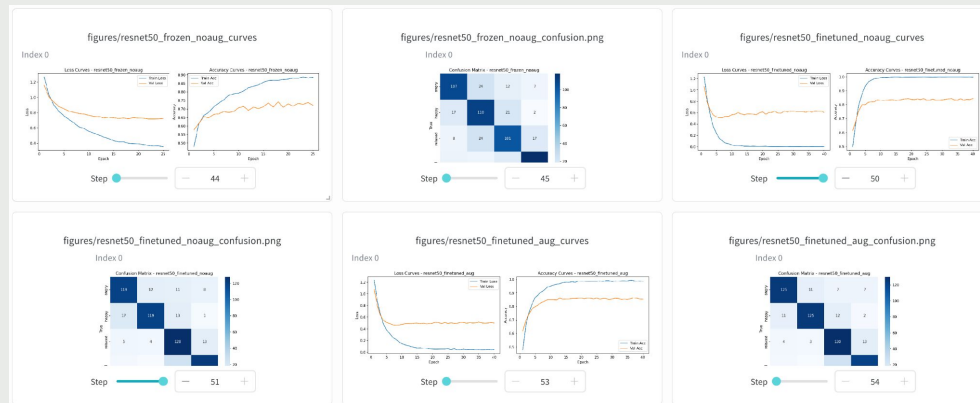
Comparative Experiments:

- Model size (ResNet-18 vs ResNet-50)
- Training strategy (frozen vs fine-tuned)
- Classifier type (linear vs MLP)

Qualitative Analysis:

- Misclassified examples
- Class-wise failure patterns

	model	fine_tuned	augmentation	accuracy	f1_score
1	resnet18	False	False	0.675	0.673
2	resnet18	True	False	0.7883	0.7884
3	resnet50	False	False	0.7267	0.7275
4	resnet50	True	False	0.8133	0.8144
5	resnet50	True	True	0.835	0.8361



Configurations

Run	Backbone	Head	Freeze	Trainable Params	Total Params
resnet18_linear_none	ResNet-18	Linear	None	11.2M	11.2M
resnet50_linear_none	ResNet-50	Linear	None	23.5M	23.5M
resnet18_mlp_all_frozen	ResNet-18	MLP	All	265K	11.4M
resnet50_mlp_all_frozen	ResNet-50	MLP	All	1.05M	24.6M
resnet15_mlp_head_layer4	ResNet-50	MLP	head->layer4	16M	24.6M

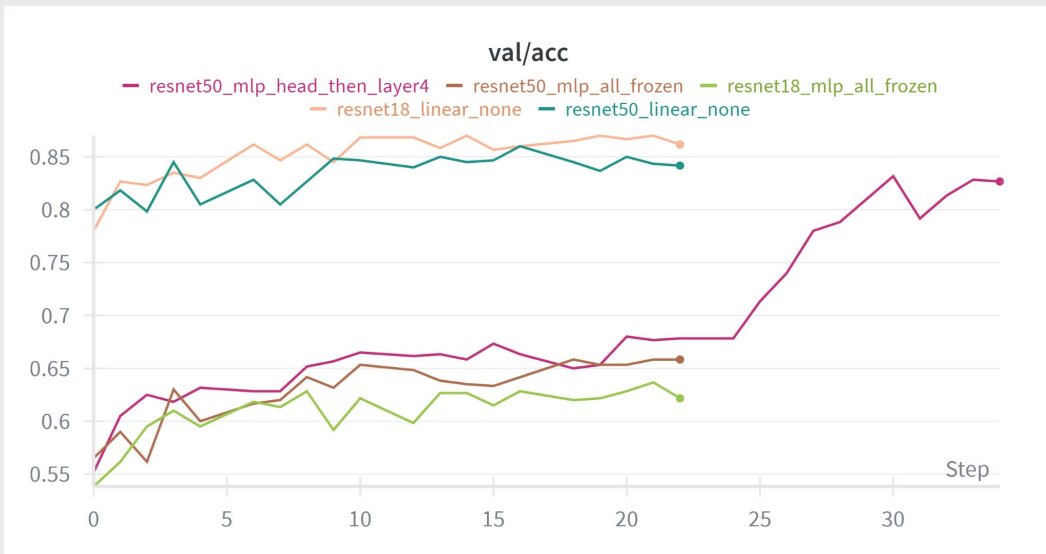
Figures



Findings:

- Resnet18_linear_none: 99.7%
- Resnet50_linear_none: 99.5%
- Resnet18_mlp_all_frozen (at epoch 22) : 72.1%
- Resnet18_mlp_all_frozen (at epoch 34) : 93.4%
- Resnet50_mlp_all_frozen: 68.9%
- Resnet18_mlp_head_layer4: 65.6%

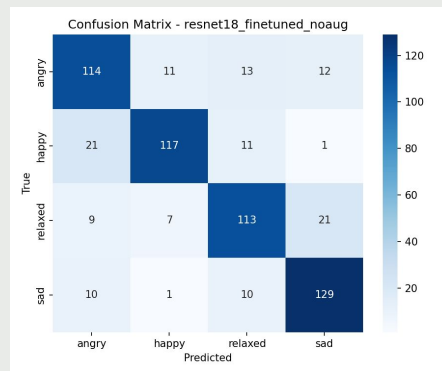
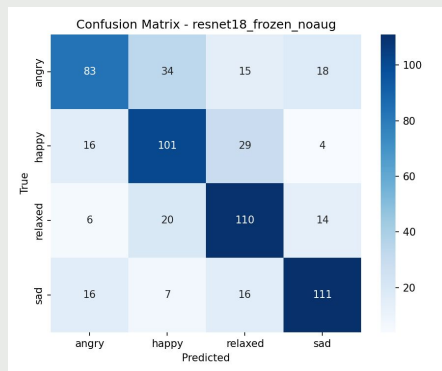
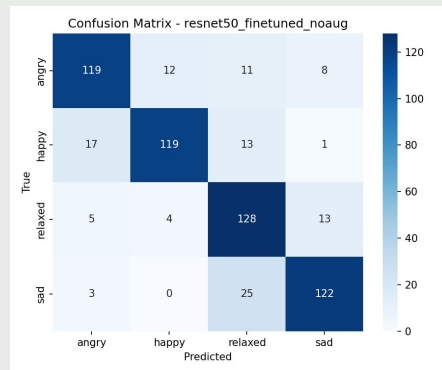
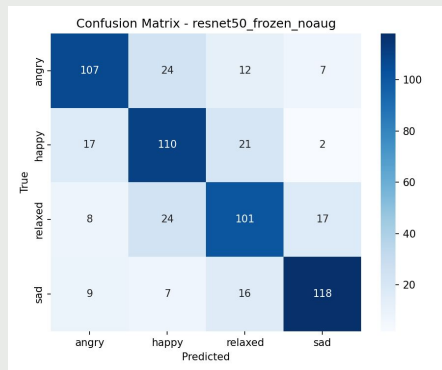
Figures



Findings:

- Resnet18_linear_none: 86.2%
- Resnet50_linear_none: 84.2%
- Resnet18_mlp_all_frozen (at epoch 22): 67.8%
- Resnet18_mlp_all_frozen (at epoch 34): 82.6%
- Resnet50_mlp_all_frozen: 65.8%
- Resnet18_mlp_head_layer4: 62.2%

Confusion Matrices



Overall Performance

- Sad best predicted (118), followed by happy (110), angry (107), relaxed (101)
- Diagonal dominates — model is generally making sensible predictions

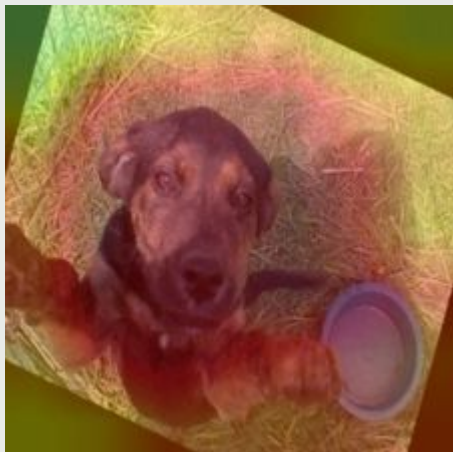
Angry → Happy (24 misclassifications)

- Both can show open mouth / teeth
- Model struggles to distinguish threat display from excited panting

Relaxed → hardest class overall

- Misclassified as happy (24), sad (17), angry (8)
- No strong unique visual marker — overlaps with every other class
- Neutral posture looks similar to mildly happy or low-energy sad

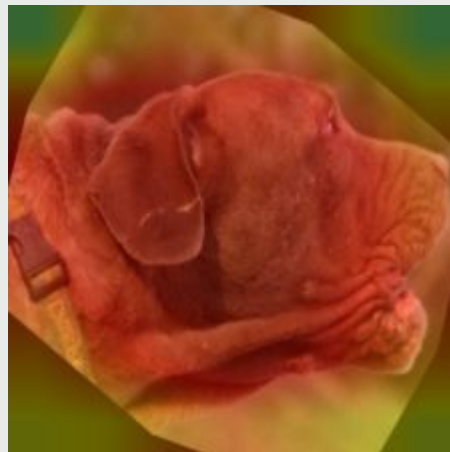
Images/Attention Map of Misclassification



True: sad | Pred: happy



True: relaxed | Pred: angry



True: angry | Pred: sad

Key Findings

Full Fine-Tuning Dominates

- ResNet-18 and ResNet-50 (no freezing) reach 84–86% val accuracy by epoch 22
- Pretrained ImageNet features alone are not enough — backbone must adapt to dog emotions

ResNet-18 Beats ResNet-50 in Full Fine-Tune

- 86.2% vs 84.2% val accuracy with less than half the parameters
- ResNet-50's extra capacity causes mild overfitting on this small dataset

Staged Unfreeze Recovers Strongly

- Starts weak (~68% val) but climbs to 82.7% after layer4 is unlocked in phase 2
- Stabilising the head before unfreezing layer4 is an effective strategy

Deeper Head Doesn't Fix Freezing

- MLP (2048→512→4) vs Linear (2048→4) makes little difference when backbone is frozen
 - Bottleneck is in the backbone features, not classifier capacity
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References

- <https://www.kaggle.com/datasets/danielshanbalico/dog-emotion>
 - <https://www.image-net.org/download.php>
 - <https://docs.pytorch.org/vision/main/models/generated/torchvision.models.resnet50.html>
 - <https://unsplash.com/>
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